CIE AS PHYSICS (9702) • QUESTION PAPER

7.1 Progressive Waves

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 20 Questions (20 Marks)

Question 1 Easy [1 mark]

A wave has a frequency of 4 GHz.

What is the period of the wave?

A 25 000 μs

B 250 ps

C 25 ns

D 2.5 ns

Question 2 Easy [1 mark]

A light with wavelength of 600 nm has an orange colour.

What is the frequency of this light?

A 500 kHz

B 500 THz

C 180 GHz

D 180 Hz

Question 3 Easy [1 mark]

A sound wave has a frequency of 50 Hz and a speed of 330 m s^{-1}

If two waves have a phase difference of 60˚, what is the possible distance between two points on the wave?

A 6.6 m

B 2.2 m

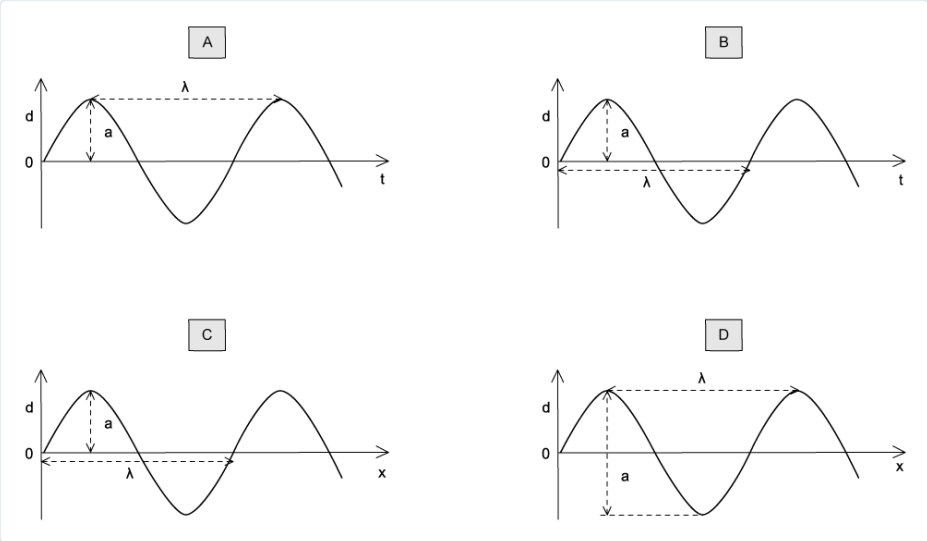
C 1.1 m

D 0.03 m

Question 4 Easy [1 mark]

A sound wave has a displacement *y* at a distance *x* from its source at time *t*.

Which graph shows correctly the amplitude *a* and the wavelength *λ* of the wave ?



A

B

C

D

Question 5 Easy [1 mark]

The following are statements about waves.

1	Longitudinal waves can be polarised.
2	The amplitude of a wave is directly proportional to the energy transferred by the wave.
3	All electromagnetic waves travel at the same speed in a vacuum.
4	The frequency of infra-red light is greater than the frequency of ultra-violet light.

Which ones are incorrect?

A 1, 3 and 4

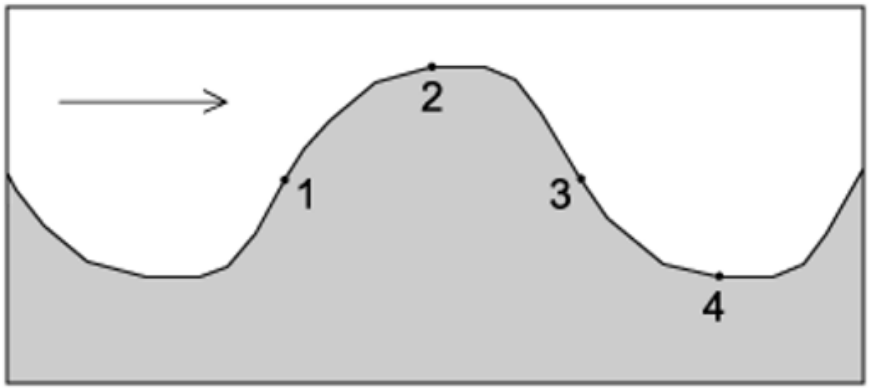
B 1, 2 and 4

C 1, 2 and 3

D 2, 3 and 4

Question 6 Easy [1 mark]

The vertical cross-section diagram shows a water wave moving from left to right.



Which points on the diagram show the water moving upwards with maximum speed?

A 1 and 3

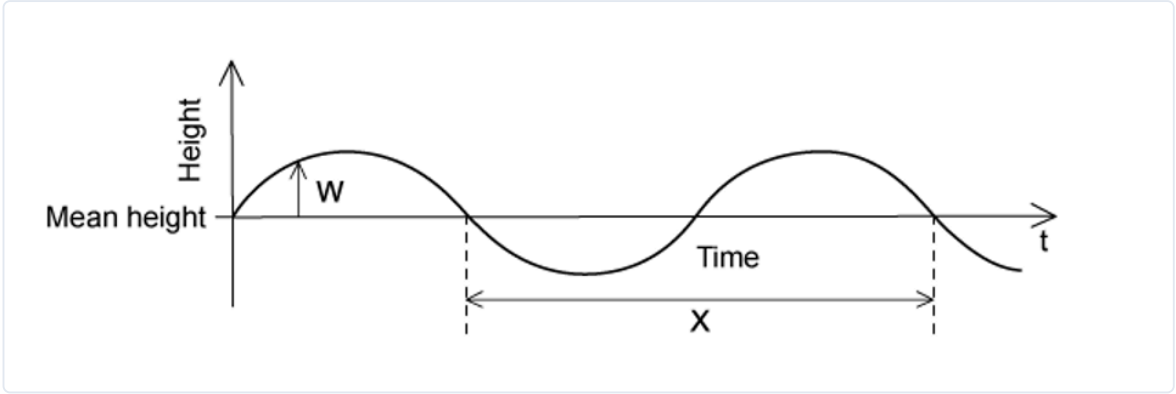
B 3 only

C 1 and 4

D 4 only

Question 7 Easy [1 mark]

The diagram below is a graph showing how the height of water at a point in a marina varies with time *t* as the waves pass the point.



What is the correct label for W and X?

	W	X
A	amplitude	period
B	displacement	wavelength
C	amplitude	wavelength
D	displacement	period

A

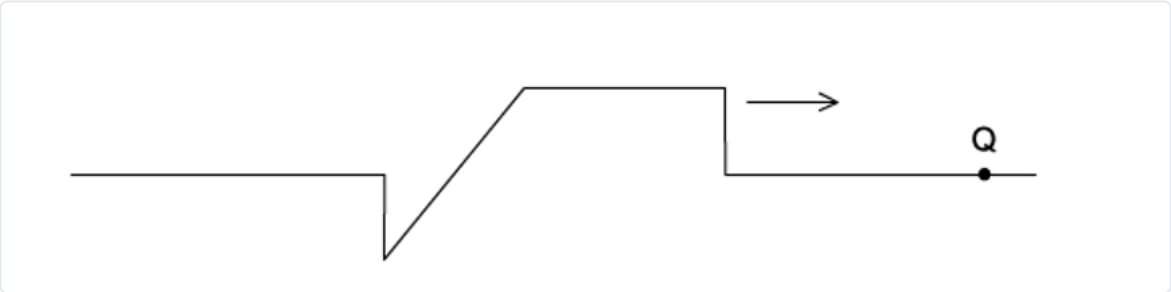
B

C

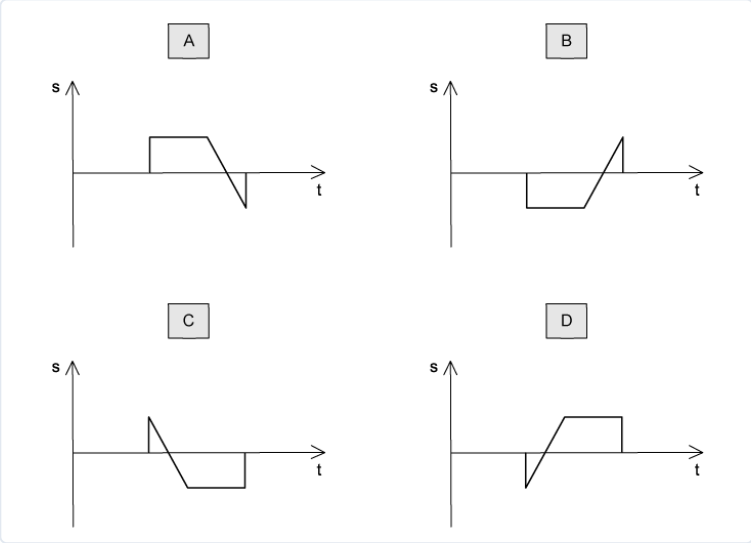
D

Question 8 Medium [1 mark]

The diagram shows a wave moving along a stretched rope in the direction shown.



Which of the following will be the correct graph to show the variation with time *t* of the displacement *s* of the particle Q in the rope ?



A

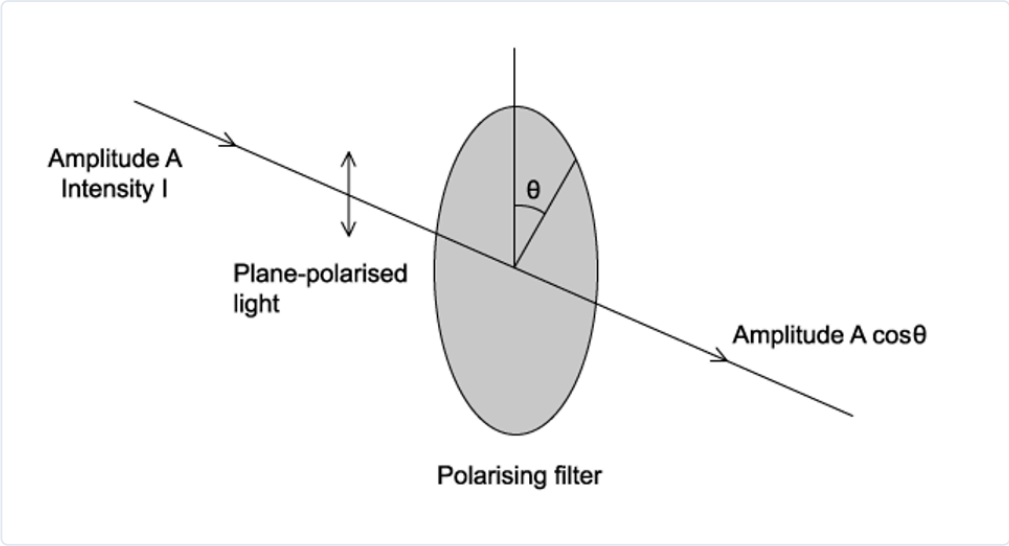
B

C

D

Question 9 Medium [1 mark]

A plane polarised light of amplitude *A* is passed through a polarising filter as show in the diagram. The emerging light has an amplitude of A cos θ



If the intensity of the beam is *I*, what is the intensity of the emerging light when θ is 60.0˚?

A 0.250 *I*

B 0.500 *I*

C 0.750 *I*

D 0.866 *I*

Question 10 Medium [1 mark]

A sound wave with an amplitude a of 0.2 mm has an intensity of 2.5 W m^{-2}

If a wave with the same frequency has an amplitude of $2a$ what is the intensity of the wave?

A 10.0 W m^{-2}

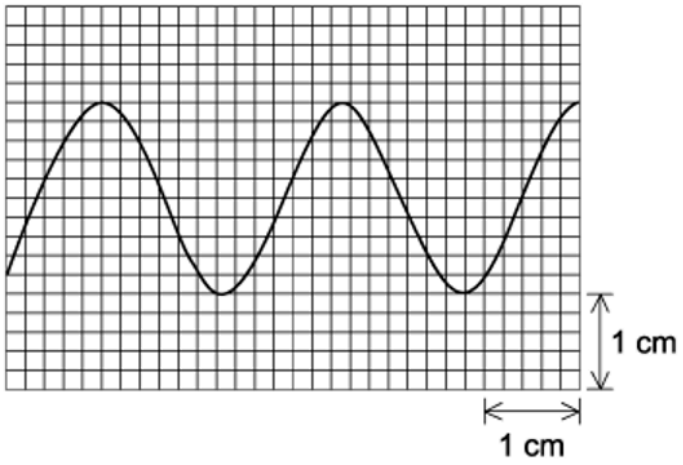
B 9.0 W m^{-2}

C 6.0 W m^{-2}

D 4.2 W m^{-2}

Question 11 Medium [1 mark]

A sound wave can be displayed on a cathode ray oscilloscope (C.R.O.). The time base is set to 5 μs mm^{-1}.



What is the frequency of the sound wave?

A 5 Hz

B 100 Hz

C 5 kHz

D 100 kHz

Question 12 Medium [1 mark]

A source of sound power P is situated in an open space. The intensity I of sound at distance r from this source is given by

$$I = \frac{P}{4\pi r^2}$$

How does the amplitude A of the vibrating air molecules vary with the distance r from the source ?

A $A \ \alpha \ \frac{1}{r}$

B $A \ \alpha \ \frac{1}{r^2}$

C $A \ \alpha \ r$

D $A \ \alpha \ r^2$

Question 13 Medium [1 mark]

Radio waves are electromagnetic waves that travel at speed c in a vacuum with a frequency f and wavelength λ.

Which row in the table describes the wavelength and speed of radio waves with a frequency $3f$?

	Speed in a vacuum	Wavelength
A	$3c$	3λ
B	c	$\frac{\lambda}{3}$
C	c	3λ
D	$\frac{c}{3}$	$\frac{\lambda}{3}$

- **A**
- **B**
- **C**
- **D**

Question 14 Hard [1 mark]

The equation below shows the speed of v waves in deep water

$$v^2 = \frac{g\lambda}{2\pi}$$

Where g is the acceleration of free fall and λ is the wavelength of the waves. The wavelength λ and the frequency f of the wave were measured.

If the graph was to give a straight line through the origin which of the following graphs should be plotted?

A f against λ^2

B f^2 against λ

C f against $\frac{1}{\lambda}$

D f^2 against $\frac{1}{\lambda}$

Question 15 Hard [1 mark]

A light wave is incident normally on a surface of area S with an amplitude A. The power per unit area reaching the surface is P.

If the light is focused on a smaller area of $\frac{1}{3}$ S, and the amplitude of wave is doubled to $2A$ what is the power per unit area?

A $6P$

B $12P$

C $18P$

D $36P$

Question 16 Hard [1 mark]

All electromagnetic waves travel at speed c in a vacuum with a frequency f and wavelength λ.

Which row in the table describes the wavelength and speed of electromagnetic waves with a frequency $\frac{f}{2}$?

	Speed in a vacuum	Wavelength
A	$2c$	2λ
B	c	2λ
C	c	$\frac{\lambda}{2}$
D	$\frac{c}{2}$	$\frac{\lambda}{2}$

A

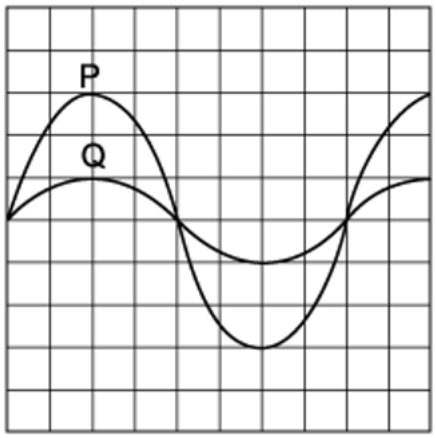
B

C

D

Question 17 Hard [1 mark]

Sound waves can be displayed on a cathode ray oscilloscope (C.R.O.). The graph below represents two waves seen on the screen. They are labelled P and Q as shown in the diagram



What is the ratio $\frac{intensity\ \ of\ \ sound\ \ wave\ \ P}{intensity\ \ of\ \ sound\ \ wave\ \ Q}$?

A $\frac{\sqrt{3}}{1}$

B $\frac{9}{1}$

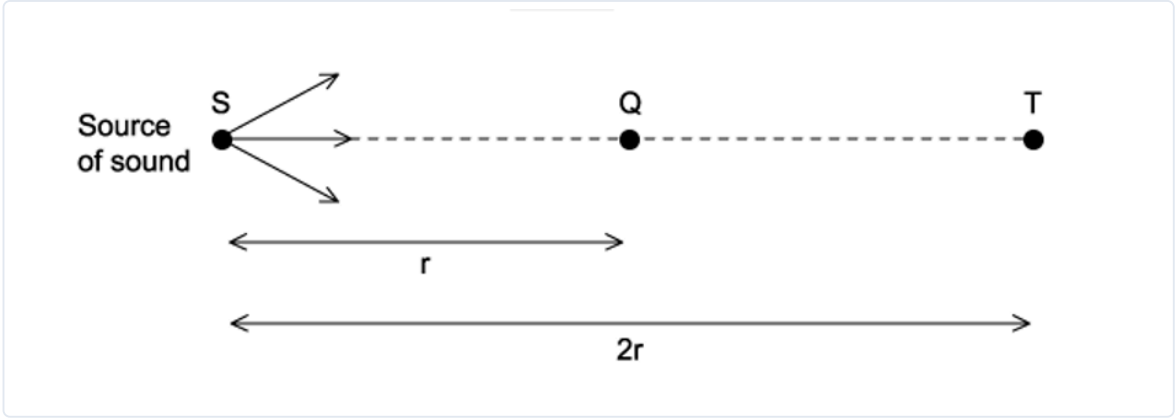
C $\frac{1}{1}$

D $\frac{3}{1}$

Question 18 Hard [1 mark]

The intensity I of a sound at point Q is inversely proportional to the square of the distance x of Q from the source of the sound. This can be shown by the equation

$$I \propto \frac{1}{x^2}$$



Molecules of air at Q are a distance r from S and oscillate with amplitude 8.0 μm.

Point T is a distance $2r$ from S.

What is the amplitude of oscillation of air molecules at T?

A 4.0 μm

B 2.0 μm

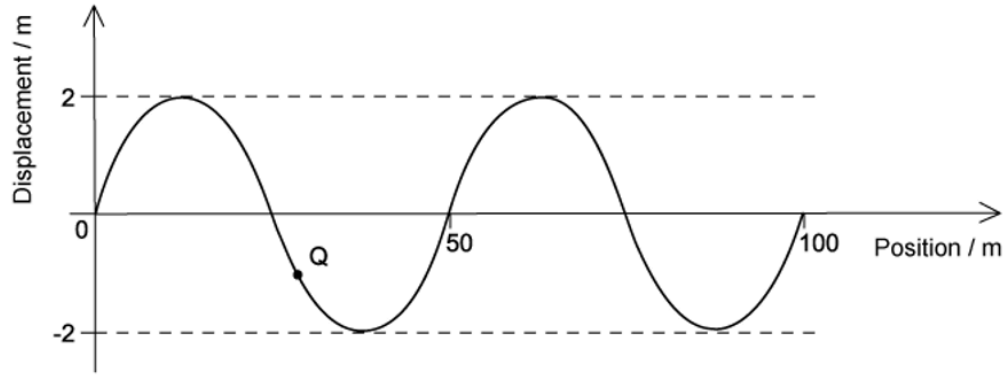
C 2.8 μm

D 1.4 μm

Question 19 Hard [1 mark]

The graph shows a sinusoidal wave in water travelling at a speed of 8.0 m s^{-1}.

The maximum speed of the particles in the wave is $2\pi af$, where a is the amplitude and f is the frequency.



An object Q is floating on the water it has a mass of 2.5 × 10^{-3} kg.

What is the maximum kinetic energy of Q due to the wave? Assume the motion is vertical.

A 4.1 mJ

B 5.1 mJ

C 39 mJ

D 64 mJ

Question 20 Hard [1 mark]

A wave has a speed of 340 m s^{-1} and a frequency of 500 Hz

If two points on a wave are 0.17 m apart, what is the phase difference?

A $\frac{\pi}{4}$ rad

B $\frac{\pi}{2}$ rad

C $\frac{3\pi}{4}$ rad

D π rad